

MANAGEMENT REVIEW

Management without theatre.

KPIS

QUALITY: FIRST PASS YIELD AND THE COST OF QUALITY

Where the idea of a cost of quality came from, how much poor quality costs, the four groups of cost, whether quality is still free, and a card for First Pass Yield.



15–20%

of sales revenue is what quality-related costs reach in many organisations, by ASQ's estimate; in some, as much as 40% of operations.

ASQ

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Every operation pays for its mistakes: the order packed twice, the delivery made again, the complaint answered. This issue asks how much of that cost an operation can see, and which measure shows it earliest.

Joseph Juran's handbook of 1951 first treated quality costs and called the cost of failures gold in the mine. Armand Feigenbaum split them into four groups in 1956; Philip Crosby argued in 1979 that quality is free and that what costs money is not doing a job right the first time. ASQ estimates that in many organisations these costs reach 15–20% of sales. First Pass Yield shows, step by step, how much work goes right the first time.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure

Every figure has its source and year at the foot of its page.

Our reading

Where the editors interpret rather than the research, it says so.

Practice

The steps and the card are proposals to try, not research results.

COVER STORY

Gold in the mine

The idea that quality has a cost that can be counted is older than most quality programmes. It took shape between 1951 and 1979, in a handbook, an article and a book.

**1951**

Juran's Quality-Control Handbook first treats quality costs and calls the cost of failures gold in the mine.

**1956**

Feigenbaum, in Harvard Business Review, splits them into four groups: prevention, appraisal, internal and external failure.

**1979**

Crosby, Quality Is Free: quality is conformance to requirements, measured by the price of nonconformance.

“What costs money are the unquality things — all the actions that involve not doing jobs right the first time.”

Juran's point was that failure costs can be cut sharply by investing in improvement, so they are a source of money, not only a loss. Crosby went further: the money is in the price of not conforming.

OUR READING

The cost of poor quality is not one line in the accounts. It is spread across rework, extra checks, returns and the time of the people who put things right.

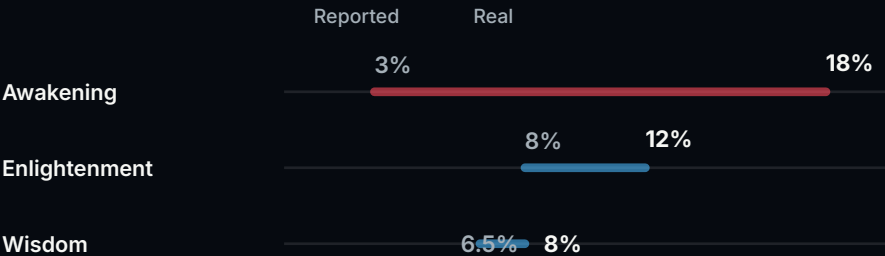
Sources: Joseph M. Juran (ed.), McGraw-Hill, 1951; Armand V. Feigenbaum, Harvard Business Review, 1956; Philip B. Crosby, McGraw-Hill, 1979

THE NUMBERS

What poor quality costs

ASQ estimates that in many organisations the true quality-related costs reach 15–20% of sales revenue, and in some as much as 40% of total operations. It gives no method: the figure is an estimate, not a measurement.

COST OF QUALITY AS A SHARE OF SALES IN CROSBY'S MATURITY GRID, BY STAGE



In his maturity grid of 1979, Crosby set the cost of quality that companies report against what he estimated it really is. In the first stage, uncertainty, the cost is not known; he put it at 20% of sales. In the last, certainty, at 2.5%.

Outside the factory the cost shows too: an OECD review of the evidence found that 15% of hospital spending and activity goes to treating failures of patient safety.

OUR READING

The reported cost of quality is what someone has counted. Most of the real cost sits where nobody counts: in rework and in the time spent fixing.

Sources: ASQ, 2026; Philip B. Crosby, McGraw-Hill, 1979; Luke Slawomirski, Ane Auraaen & Niek Klazinga, OECD, 2017

Crosby's figures are illustrative estimates from 1979, not measured data.

THE MODEL

Four groups of cost

The model Feigenbaum described in 1956 is still the most used: two groups of cost are spent to get quality right, two are paid when it goes wrong.

The cost of good quality

Prevention: training, process design, keeping the quality system

Appraisal: measuring, inspecting, testing, auditing

The cost of poor quality

Internal failure, found before the customer: scrap, rework, failure analysis

External failure, found by the customer: returns, warranty, complaints

HYPOTHETICAL EXAMPLE, A WAREHOUSE

Prevention	Training on the new labels
Appraisal	Scanning every pallet before loading
Internal	Repacking a wrong order before it leaves
External	A second delivery after a complaint

The later a mistake is found, the more steps and people it has already passed. The examples are invented.

OUR READING

A team that counts only failures sees half of the model. The other half is where the money for fewer failures is spent.

Sources: Armand V. Feigenbaum, Harvard Business Review, 1956; ASQ, 2026

The four groups are Feigenbaum's, as ASQ describes them today; the examples are the editors'.

[More in the essay](#)

KPIs the team trusts

WHAT THE RESEARCH SAYS

Is quality still free?

Two studies in the International Journal of Quality & Reliability Management look at how the cost of quality is used, and at what happens to it as quality improves.

01 Most programmes do not use it

Reviewing the research on cost-of-quality models, Schiffauerova and Thomson found that most quality programmes do not use the cost of quality. Prevention-appraisal-failure is the most used model, and companies that use one report lower quality costs and better quality for their customers.

02 As quality rises, total cost falls

With data from manufacturing, Plewa, Kaiser and Hartmann found that as quality performance rises, the total cost of quality and the cost of failures fall, while spending on prevention and appraisal does not rise significantly.

OUR READING

The research supports Crosby more than it contradicts him: better quality went with lower total cost. It shows a link, not a recipe.

Sources: A. Schiffauerova & V. Thomson, International Journal of Quality & Reliability Management, 2006; M. Plewa, G. Kaiser & E. Hartmann, International Journal of Quality & Reliability Management, 2016

Both rest on reported cases or correlations; we could not see the sample size of the 2016 study.

HOW IT IS MEASURED

First Pass Yield, step by step

First Pass Yield is the share of units that pass a step the first time, without rework or scrap. Multiplied across all the steps, it gives the Rolled Throughput Yield: the chance that a unit passes the whole process without a defect.

1

List the steps

From order to delivery, in the order the work happens.

2

Count the first-time passes

A unit that was reworked does not count as good.

3

Multiply

Four steps at 85% each give 52.2%: about half the units pass every step the first time.

4

Start with the lowest step

Improve the step with the lowest yield first.

HYPOTHETICAL EXAMPLE, AN ORDER THROUGH A WAREHOUSE

Picking 97 of 100 right the first time

Packing 99 of 100

Loading 98 of 100

Delivery 96 of 100

Each step looks good, yet only 90.3% of orders pass all four the first time. The numbers are invented.

Source: iSixSigma, 2026

The steps and the example are the editors'; the formula follows iSixSigma.

TOOL OF THE ISSUE

The First Pass Yield card

One card for one process. Fill in a week of counts, multiply, and choose the step to improve first.

01 Process from where to where, and which unit is counted

02 Steps and First Pass Yield step · units in · right the first time · share

03 Rolled Throughput Yield the shares of all the steps, multiplied

04 Weakest step the lowest share, and its most frequent failure

05 Where it is paid internal failure or external failure

06 One change what we change this week, and when we count again

Sources: Armand V. Feigenbaum, Harvard Business Review, 1956; iSixSigma, 2026

A practice proposed by the editors, on Feigenbaum's groups of cost and the yields as iSixSigma defines them.

Open the tool: [Damage Control](#)

stivencatalyst.com/tools/damage-control/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

Quality-Control Handbook

Joseph M. Juran (ed.), McGraw-Hill, 1951

Total Quality Control

Armand V. Feigenbaum, Harvard Business Review, 1956

Quality Is Free: The Art of Making Quality Certain

Philip B. Crosby, McGraw-Hill, 1979

What is Cost of Quality (COQ)?

ASQ, 2026
<https://asq.org/quality-resources/cost-of-quality>

**The economics of patient safety:
Strengthening a value-based approach to
reducing patient harm at national level**

Luke Slawomirski, Ane Auraaen & Niek Klazinga, OECD, 2017
<https://doi.org/10.1787/5a9858cd-en>

**A review of research on cost of quality
models and best practices**

A. Schiffauerova & V. Thomson, International Journal of Quality & Reliability Management, 2006
<https://doi.org/10.1108/02656710610672470>

**Is quality still free? Empirical evidence on
quality cost in modern manufacturing**

M. Plewa, G. Kaiser & E. Hartmann, International Journal of Quality & Reliability Management, 2016
<https://doi.org/10.1108/IJQRM-11-2014-0189>

What is rolled throughput yield?

iSixSigma, 2026
<https://www.isixsigma.com/dictionary/rolled-throughput-yield-rtv/>

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked “via”. The editors' interpretation is marked “Our reading”. Figures that could not be confirmed are not in the issue.



MANAGEMENT REVIEW

Management without theatre.

Every issue, one management question, checked against the best research.

ALL ISSUES

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